TOLUWANI ERIOLUWA ABRAHAM

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PROFESSIONAL SUMMARY

High-agency Computer Science graduate (B.Sc. Hons, Covenant University) and Systems & Smart Contract Engineer specializing in **decentralized settlement protocols (EVM)**, **order-book market data architectures**, and **financial risk engineering**. Architect of D-MEX (trustless atomic swap engine on Base) and STELCERA Terminal (low-latency multi-desk market analytics). Track record of delivering measurable outcomes: **38% on-chain gas reductions**, **120 FPS canvas tick renders**, **cryptographic payment idempotency**, and **enterprise corporate activations**.

TECHNICAL EXPERTISE

Exchange & Settlement:	Order Book Mechanics, Liquidity Aggregation, Market Making Logic, Idempotent Settlement, Double-Entry Ledgering
Blockchain & Smart Contracts:	Solidity, Inline Yul Assembly, Base (EVM), HTLC Atomic Swaps, ERC-20/721/1155, UUPS Proxies, Foundry/Forge, IPFS
Distributed Backend:	Python, C++, TypeScript, Go, Node.js, FastAPI, Apache Kafka, Redis (Pub/Sub & Caching), PostgreSQL, REST APIs, Docker
Markets & Quantitative:	Order Book Depth & Spread Analytics, Slippage Modeling, Volatility Indices, Risk Management, TradingView Charts, MT5

SELECTED ENGINEERING & WEB3 SYSTEMS

D-MEX Protocol — Decentralized Atomic Barter & Settlement Engine **2026**

Solidity, Base (migrated from Scroll Sepolia), Yul Assembly, UUPS Proxies, Foundry *GitHub: abraham-DT/DMEX-Protocol*

- **Eliminated counterparty risk & front-running:** Architected a zero-custody atomic swap protocol bridging ERC-20 and ERC-1155 tokens on Base using Hashed Timelock Contracts (HTLCs) with asymmetric safety windows ($T_1 > T_2$) and an off-chain Python guardian monitoring pending mempool transactions.
- **Cut on-chain gas costs by 38%:** Optimized low-level storage packing and bytecode using inline Yul assembly, verifying contract state invariants with **0 security vulnerabilities** across automated Foundry invariant fuzzing test suites.

STELCERA Terminal — High-Frequency Market Analytics & Order-Book Workstation **2026**

Next.js 15, React 19, FastAPI, WebSockets, HTML5 Canvas, TradingView Lightweight Charts *Production Deployment*

- **Sustained smooth 120 FPS tick renders:** Streamed real-time order-book depth and spreads across crypto/FX pairs with sub-50ms latency via asynchronous FastAPI WebSockets, drawing directly onto HTML5 Canvas via `requestAnimationFrame` to eliminate React DOM reconciliation overhead.
- **Enabled instant portfolio risk hedging:** Engineered a mathematical pricing engine calculating real-time Black-Scholes analytical values and full Greeks sensitivities (Δ, Γ, ν, Θ, ρ) for live multi-asset portfolios.

REAVO — Student Electronics Store & Inventory Management Dashboard **2026**

Node.js, Express, Vite / React, Supabase (PostgreSQL), KoraPay API, Tailwind CSS *Live Production (reavo-app.vercel.app)*

- **Eliminated payment fraud & double-crediting:** Implemented cryptographic HMAC SHA-256 webhook validation with atomic idempotency locks on KoraPay checkouts across student gadget retail.
- **Prevented inventory overselling & data leaks:** Architected Supabase PostgreSQL Row Level Security (RLS) policies and automated hourly stock audit crons ensuring 100% warehouse catalog synchronization.

CivOS — Virtual World Simulation Architecture & Deterministic State Validation **2026**

C++, Apache Kafka, ERC-4337, Redis, Vector DB *System Research Prototype*

- **Eliminated AI game-rule hallucinations & API insolvency:** Built a deterministic C++ state validation engine enforcing physics, collision, and inventory rules before committing state, isolating LLMs strictly to dialogue.

SIPHOON OS — Quantitative Trading Journal & Risk Engine **2026**

Next.js 15, TypeScript, PostgreSQL, Prisma, Statistical Modeling *Fintech Architecture*

- **Surfaced hidden execution slippage & edge decay:** Automated reconciliation between trade execution logs and trading journals, paired with dynamic drawdown tracking and R-multiple statistical distributions.

PROFESSIONAL EXPERIENCE

Kucheza Gaming — Industrial Training Fellow (Spatial R&D & Systems) **Apr 2025 – Oct 2025**

Esports Operations, Enterprise Decks & Educational Technology *Lagos, Nigeria*

- **Secured enterprise brand activations for MTN, Rexona, Fan Milk, and Infinix:** Co-designed spatial 3D tournament arena staging layouts and technical corporate sponsorship proposals for broadcast esports.
- **Achieved a 92% student completion rate:** Led educational curriculum engineering for **CodePlay Naija**, instructing secondary school students in programming logic, algorithms, and computational thinking.

Blueshels Ltd — Web Designer & Frontend UI Engineer **2024 – 2025**

Web Architecture, Asset Optimization & Conversion Pipelines *Lagos, Nigeria*

- **Boosted client checkout conversions by 28%:** Engineered responsive web interfaces, structured conversion funnels, and accessible UI systems compliant with WCAG 2.1 accessibility standards.
- **Achieved sub-second page loads:** Refactored frontend asset delivery pipelines and optimized critical rendering paths to deliver 95+ Lighthouse performance scores.

EDUCATION & CREDENTIALS

Covenant University — B.Sc. (Hons) in Computer Science **2022 – 2026**

CGPA: 3.77 / 5.00 (Second Class Upper Division) *Ota, Ogun State, Nigeria*

- **Final Year Thesis:** *Developing an Atomic Swap Protocol to Guarantee Zero Counterparty Risk in Cross-Game Digital Asset Exchange* (Dr. Azu Ezenwoke).
- **Relevant Coursework:** Distributed Systems, Applied Cryptography, Advanced Data Structures, Operating Systems, Systems Programming.
- **Certifications:** Prompt Engineering & Large Language Models — OBTranslate.